



Block screenshots in WhatsApp

It looks like WhatsApp is finally getting a much awaited screenshot blocking feature. <https://digit.in/oct22-33>



Twitter Edit feature is here

Twitter's Edit feature has finally made it to the app. It is only available to Twitter Blue members. <https://digit.in/oct22-34>

AMD Ryzen 9 7950X Desktop Processor

Knocking it out of the park, yet again

AMD Ryzen 7950X, like its predecessor, the AMD Ryzen 9 5950X, is a 16-Core / 32-Thread processor featuring a host of new improvements including a new microarchitecture in the form of AMD Zen 4, much higher clock speeds, a new AM5 LGA socket, and much more. With so many high-impact changes over the Ryzen 5000 series processors, the new AMD Ryzen 7000 series of processors has plenty of things that can really push it over the edge but it has an equal number of things that can go wrong. Hopefully, that's not the case.

SPECIFICATIONS

Moving to the more power-efficient TSMC 5nm process is going to help the Ryzen 7000 quite a bit on the power efficiency front. The Ryzen 7000 series can eke out up to 49 per cent more performance on a 16-Core processor compared to the Ryzen 5000 series at the same socket power and core count. And if the performance were to be normalised, then the Ryzen 7000 series can manage to do so while consuming 62 per cent less power. Obviously, these improvements are not just because of the shift to the TSMC 5nm process but also because of all the architectural improvements. The AMD Ryzen 9 7950X has a base clock speed of 4.5 GHz and a Turbo clock speed of 5.7 GHz. And this is without an improvement to the Precision Boost algorithm.

As for the other specifications, the L3 Cache per core has remained the same but the L2 Cache has been doubled and is private to each core. L1 Cache is also the same with 32K for the Instructions and 32K for data. The TDP has been raised to 170W compared to the 5950X which was designed for 105W. Always check the peak parameters of the motherboard's VRM before you decide on



buying one. Lastly, all processors falling under the Ryzen 7000 family will have inbuilt graphics capabilities and will get RDNA 2 IGP Cores in all four processors.

PERFORMANCE

Starting with Cinebench R23 which has been the mainstay of most CPU benchmarks. The AMD Ryzen 9 7950X clearly beat the Intel Core i9-12900K in both the single-threaded as well as the multi-threaded benchmarks with scores of 1997 and 14922, respectively. The 12900K had scored 1971 and 25975, respectively. In Blender, we saw the 7950X rendering the test scenes way faster than the 12900K with a performance difference of 67-77 per cent across three tests. Moving to the IGP and the ISA of the CPUs. Intel has several accelerators leveraging their Xe GPU architecture and a lot of algorithms are already quite optimised for it. As a result, we see the 12900K doing much better than the Ryzen 9 7950X in the AI benchmark. Talking about RAM, Intel has XMP memory profiles and now AMD has EXPO. To see how the memory controllers handle the processor without either profiles being set, we manually set the modules to 5200 MT/s and the

sub-timings were left to the board to auto-configure. What we observed was that when the modules are tuned, both processors manage to get about 65 ns as the memory latency. However, when left to the board to auto-configure, the AMD system got a slightly lower latency. From the 1080p gaming benchmarks, we saw that it was at par with the 12900K in most benchmarks with the exception of one game — Far Cry 6 — where the Intel processor maintained a slight lead. As for HITMAN 3 and Shadow of the Tomb Raider, the Zen 4 processor seems to have gained a significant lead of up to 40 FPS. During a sustained 10 min heavy workload, the CPU Package power averaged around 210 W while hitting a peak of 223 W. In the same heavy load test, we also logged the temperature data. The Ryzen



PERFORMANCE..... 90
VALUE FOR MONEY...75

being a 16-Core processor has two Core Complex Dies (CCD) and they both logged slightly different temperatures with CCD1 averaging a little higher at 93.4 degrees Celsius while CCD2 averaged at 87.4 degrees Celsius. All cores put together, averaged at 85 degrees Celsius. As for peak temperatures, CCD1 hit 96.4 degrees and CCD2 hit 91.4 degrees, respectively.

VERDICT

The AMD Ryzen 9 7950X with its new Zen 4 architecture easily bests the Intel Core i9-12900K in most benchmarks. Even in gaming, it equals or bests the 12900K in most benchmarks at 1080p. The inbuilt RDNA 2 graphics cores have their work cut out for them to match Intel's Xe in terms of AI workloads. For the price of approximately ₹70,000, the AMD Ryzen 9 7950X is one of the prime candidates for an enthusiast gaming and content creation rig.

—Mithun Mohandas

SPECIFICATIONS

CORES: 16 | THREADS: 32 | BASE CLOCK: 4.5 GHz | BOOST CLOCK: 5.7 GHz | PROCESS NODE: 5nm | L3 CACHE: 64 MB | L2 CACHE: 512 KB per core | TDP: 170 W.

CONTACT

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